Global Genetics MX - Premium POS System

A professional, mobile-first Point of Sale (POS) system built for Global Genetics MX with Supabase backend integration, advanced PDF invoice generation, and customer management capabilities.

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🎯 Overview

Global Genetics MX POS is a comprehensive point-of-sale solution designed specifically for the Global Genetics MX company. The system manages product inventory, customer relationships, sales transactions, and generates professional PDF invoices with company branding. Built with a mobile-first approach, it provides an intuitive interface optimized for both desktop and mobile devices.

Key Technologies:

- **Frontend:** HTML5, CSS3, JavaScript (ES6+)
- **Backend:** Supabase (PostgreSQL database with real-time capabilities)
- **Charts:** Chart.js for analytics and inventory visualization
- **PDF Generation:** jsPDF with autotable plugin
- **Design:** Premium mobile-first UI with Global Genetics MX branding

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✨ Features

Core POS Functionality

- **Dashboard (Inicio):** Real-time inventory analytics with value calculations, low stock alerts, and interactive charts
- **Sales Module (Venta):** Complete point-of-sale interface with product search, shopping cart, and payment method selection
- **Inventory Management (Stock):** Product listing, stock tracking, restock functionality, and search/filter capabilities
- **Customer Management (Clientes):** Customer registration with CFDI usage classification and welcome email functionality

Advanced Features

- **Supabase Integration:** All data persisted in Supabase PostgreSQL database with row-level security
- **Payment Methods:** Support for Efectivo, Crédito, Débito, Transferencia, and Cheque
- **CFDI Classification:** Customer records include CFDI usage types (G03, G01, G02, S01)
- **PDF Invoice Generation:** Professional invoices with company logo, itemized details, and financial summary
- **Post-Sale Actions:** Email, PDF download, and WhatsApp sharing options
- **Company Branding:** Global Genetics MX logo and colors throughout the application
- **Responsive Design:** Optimized for mobile, tablet, and desktop screens

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📁 Project Structure

Plain Text

```
global-genetics-pos/
├── pos-index.html              # Main POS application HTML
├── pos-styles.css              # Premium CSS with branding
├── pos-app.js                  # Main application logic with Supabase integrat
├── registration-landing.html   # Customer registration landing page
├── registration-styles.css     # Registration form CSS
├── registration-app.js         # Registration form logic
├── registration-thankyou.html  # Thank you page after registration
├── supabase-setup-guide.md     # Detailed Supabase backend setup instructions
```

```
├── README.md                    # This file
└── .env.example                 # Environment variables template
```

🚀 Installation & Setup

Prerequisites

- A modern web browser (Chrome, Firefox, Safari, or Edge)
- A Supabase account (free tier available at https://supabase.com)
- Git for version control
- A GitHub account for hosting the registration form

Step 1: Clone or Download the Repository

Bash

```bash
git clone https://github.com/007leading-source/global-genetics-pos.git
cd global-genetics-pos
```

Step 2: Set Up Supabase Backend

Follow the detailed instructions in `supabase-setup-guide.md` to:

1. Create a Supabase project
2. Create database tables (products, customers, sales, sale_items, stock_history)
3. Configure Row-Level Security (RLS) policies
4. Obtain API credentials

Step 3: Configure Environment Variables

Create a `.env.local` file in your project root with the following variables:

Plain Text

```
# Supabase Configuration
VITE_SUPABASE_URL=https://your-project.supabase.co
VITE_SUPABASE_ANON_KEY=your-anon-key-here

# Application Configuration
VITE_APP_TITLE=Global Genetics MX POS
VITE_APP_LOGO=https://your-cdn-url/logo.png
```

```
VITE_COMPANY_NAME=Global Genetics MX
VITE_COMPANY_RFC=ABC123456XYZ
VITE_COMPANY_ADDRESS=Jalostotitlán, Jalisco, México
VITE_COMPANY_PHONE=+52 (431 ) 101-0504
VITE_COMPANY_EMAIL=info@globalgenetics.mx

# Registration Form URL
VITE_CUSTOMER_REGISTRATION_URL=https://your-github-pages-url/registration-landi
```

Step 4: Upload Company Logo

1. Upload your Global Genetics MX logo to a CDN (e.g., Cloudinary, AWS S3, or GitHub)
2. Update the `VITE_APP_LOGO` environment variable with the CDN URL

Step 5: Test Locally

To test the POS system locally, you can use a simple HTTP server:

Bash

```bash
# Using Python 3
python -m http.server 8000

# Using Node.js (if installed )
npx http-server

# Using PHP
php -S localhost:8000
```

Then open `http://localhost:8000/pos-index.html` in your browser.

🗄️ Supabase Backend Configuration

Database Schema Overview

The system uses five main tables:

Table	Purpose	Key Fields
products	Product inventory	product_id, name, cost_price, sale_price, stock, min_stock
customers	Customer records	customer_id, name, email, phone, cfdi_usage

sales	Sales transactions	sale_id, customer_id, subtotal, discount, iva, total, payment_method
sale_items	Individual items in sales	sale_id, product_id, quantity, unit_price, total
stock_history	Audit trail for stock changes	product_id, quantity_change, reason

Row-Level Security (RLS) Policies

All tables have RLS enabled with the following policies:

- **Public Read:** All authenticated users can read all data
- **Authenticated Insert:** Only authenticated users can insert new records
- **Authenticated Update:** Only authenticated users can update existing records

For detailed setup instructions, refer to `supabase-setup-guide.md` .

🔐 Environment Variables

Variable	Description	Example
`VITE_SUPABASE_URL`	Supabase project URL	`https://xxxxxxxxxxx.supabase.co`
`VITE_SUPABASE_ANON_KEY`	Supabase anonymous key	`eyJhbGc...`
`VITE_APP_TITLE`	Application title	`Global Genetics MX POS`
`VITE_APP_LOGO`	Company logo URL	`https://cdn.example.com/logo.png`
`VITE_COMPANY_NAME`	Company name	`Global Genetics MX`
`VITE_COMPANY_RFC`	Company RFC	`ABC123456XYZ`
`VITE_COMPANY_ADDRESS`	Company address	`Jalostotitlán, Jalisco, México`
`VITE_COMPANY_PHONE`	Company phone	`+52 (431 ) 101-0504`
`VITE_COMPANY_EMAIL`	Company email	`info@globalgenetics.mx`

`VITE_CUSTOMER_REGISTRATION_URL`	Registration form URL	`https://your-github-pages-url/...`

📤 Deployment to GitHub Pages

Step 1: Create a GitHub Repository

1. Go to https://github.com/new
2. Create a new repository named `global-genetics-pos`
3. Clone the repository to your local machine

Step 2: Push Files to GitHub

```bash
git add .
git commit -m "Initial commit: Global Genetics MX POS system"
git push origin main
```

Step 3: Enable GitHub Pages

1. Go to your repository settings
2. Navigate to **Pages** section
3. Select **main** branch as the source
4. GitHub will provide a URL like `https://your-username.github.io/global-genetics-pos/`

Step 4: Access the POS System

Open `https://your-username.github.io/global-genetics-pos/pos-index.html` in your browser.

Step 5: Deploy Registration Form Separately (Optional)

For the customer registration form, you can create a separate repository:

1. Create a new repository named `global-genetics-registration`
2. Push the registration files (`registration-landing.html` , `registration-styles.css` , `registration-app.js` , `registration-thankyou.html`)
3. Enable GitHub Pages for this repository

4. Update `VITE_CUSTOMER_REGISTRATION_URL` to point to this new URL

📚 Module Documentation

Inicio (Dashboard)

The dashboard provides a comprehensive overview of business metrics:

- **Inventory Value:** Total value of all products in stock (calculated as stock × cost price)
- **Low Stock Count:** Number of products below minimum stock threshold
- **Out of Stock Count:** Number of products with zero stock
- **Inventory Alerts:** Real-time alerts for low and out-of-stock items
- **Interactive Charts:** Toggle between investment value and stock quantity views

Chart Optimization: The chart displays the top 8 products by investment value. When you have 100+ products, the chart automatically filters to show the most significant items, preventing overcrowding on mobile screens.

Venta (Sales)

The sales module enables complete point-of-sale transactions:

1. **Select Customer:** Choose from existing customers or "Público General" for walk-in sales
2. **Select Payment Method:** Choose from Efectivo, Crédito, Débito, Transferencia, or Cheque
3. **Search & Add Products:** Use the search bar to find products and add them to the cart
4. **Manage Cart:** Adjust quantities or remove items
5. **Apply Discount:** Enter a discount amount (optional)
6. **Complete Sale:** Finalize the transaction

Post-Sale Actions:

- Confirmation modal displays the invoice number
- Email option to send invoice (implementation required)
- PDF download for immediate printing
- WhatsApp sharing for customer communication

Stock (Inventory)

The inventory module manages product information and stock levels:

- **View Products:** Sortable table with product details
- **Search & Filter:** Find products by name, brand, or category
- **Restock:** Add inventory with new cost price
- **Stock Badges:** Visual indicators for stock status (OK, Low, Out)
- **Sort Functionality:** Click column headers to sort by product name or stock level

Clientes (Customers)

The customer management module handles customer registration and communication:

- **Register Customers:** Add new customers with complete information
- **CFDI Classification:** Assign CFDI usage type for tax purposes
- **Welcome Email:** Send registration link via WhatsApp
- **Customer Records:** All customer data stored in Supabase for future reference

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🔌 API Reference

Supabase Client Initialization

JavaScript

```javascript
import { createClient } from '@supabase/supabase-js';

const supabase = createClient(
    import.meta.env.VITE_SUPABASE_URL,
    import.meta.env.VITE_SUPABASE_ANON_KEY
);
```

Key Functions

Load Products

JavaScript

```javascript
async function loadProducts() {
    const { data, error } = await supabase
        .from('products')
        .select('*')
        .order('name', { ascending: true });
```

```
    return data;
}
```

Create Sale

JavaScript

```javascript
async function createSale(saleData) {
    const { data: sale, error } = await supabase
        .from('sales')
        .insert([{
            sale_id: `INV-${Date.now()}`,
            customer_id: saleData.customerId,
            subtotal: saleData.subtotal,
            discount: saleData.discount,
            iva: saleData.iva,
            total: saleData.total,
            payment_method: saleData.paymentMethod
        }])
        .select();
    return sale[0];
}
```

Add Customer

JavaScript

```javascript
async function addCustomer(customerData) {
    const { data, error } = await supabase
        .from('customers')
        .insert([{
            customer_id: `CUST-${Date.now()}`,
            name: customerData.name,
            email: customerData.email,
            cfdi_usage: customerData.cfdiUsage
        }])
        .select();
    return data[0];
}
```

🛠️ Troubleshooting

Issue: "Failed to fetch" or CORS errors

Solution: Ensure your domain is added to the CORS whitelist in Supabase Settings → API.

Issue: "Row-level security violation"

Solution: Verify that RLS policies are correctly configured in Supabase. Check that your authentication context matches the policy conditions.

Issue: Logo not displaying

Solution: Verify that the CDN URL in `VITE_APP_LOGO` is correct and accessible. Check browser console for 404 errors.

Issue: Charts not rendering

Solution: Ensure Chart.js library is loaded correctly. Check browser console for JavaScript errors.

Issue: PDF download not working

Solution: Verify that jsPDF and jsPDF-autotable libraries are loaded. Check browser console for errors.

Issue: Supabase connection errors

Solution:

1. Verify `VITE_SUPABASE_URL` and `VITE_SUPABASE_ANON_KEY` are correct
2. Check that your Supabase project is active
3. Ensure tables are created and RLS policies are enabled
4. Test connection using the Supabase dashboard

📞 Support & Contact

For issues, questions, or feature requests:

- **Email:** info@globalgenetics.mx
- **Phone:** +52 (431) 101-0504
- **GitHub Issues:** Report bugs on the project repository

📄 License

This project is proprietary software for Global Genetics MX. All rights reserved.

🙏 Acknowledgments

Built with premium design principles and modern web technologies to provide a professional point-of-sale experience for Global Genetics MX.

Technologies Used:

- Supabase for backend infrastructure
- Chart.js for data visualization
- jsPDF for invoice generation
- Font Awesome for icons
- Google Fonts for typography

📋 Version History

Version	Date	Changes
1.0.0	2026-04-16	Initial release with Supabase integration, PDF invoicing, and premium UI

Last Updated: April 16, 2026

Maintained by: Manus AI